

Computational spatial planning and policy report

Statewide secondary education accessibility suite

ODISHA SPATIAL SCHOOL EDUCATION
MASTERPLAN (2026-2031)

Operations Research Optimization, 314-Block GIS Analysis, PWD Hill Cost Calibrations,
and Multi-Tier Infrastructure Investment Architecture

Total capital outlay

₹6932 Cr

Secondary & Transit Outlay

Physical campuses

3704

2340 Upgrades + 1364 Greenfield

GSDP economic return

0.77x BCR

₹5323 Cr Direct NPV Addition

Spatial challenge and problem statement

This technical report establishes the spatial allocation architecture for achieving universal secondary education access across the State of Odisha by 2031. Under the Right to Education (RTE) Act of 2009, primary schooling saturation reached 69.9% baseline coverage within a 1 km radius. However, secondary education (Grades 9-10, 5 km norm) was not expanded at parity, creating an acute geographic transition barrier.

Across the Eastern Ghats and forested corridors, adolescent students—predominantly females—are forced to walk over 7 to 10 kilometers across steep, unpaved terrain to access the nearest high school. This logistical friction produces a severe drop in continuation rates between Grade 8 and Grade 9. Traditional demand-based funding favored politically active plain districts while leaving tribal belts unserved. This masterplan replaces subjective approvals with an objective spatial optimization model.

Algorithmic model and operational solution

The allocation framework formulates school placement as a Submodular Maximum Coverage Location Problem (MCLP) bounded by the Nemhauser-Wolsey $(1 - \frac{1}{e}) \approx 63.2\%$ approximation guarantee. The algorithm tabulates unserved habitations across all 314 CD blocks and applies Tobler hiking friction curves to model realistic pedestrian travel speeds across terrain gradients.

Civil engineering estimates incorporate Public Works Department (PWD) hill area cost multipliers (+18% to +30%) to reflect actual material transport expenses. Where low habitation density makes brick-and-mortar schools economically unviable, the model deploys a synchronized transit fleet of 2250 mini-buses and 1308 feeder vans alongside 743 dedicated girls' residential hostels, lifting statewide secondary coverage from 52.3% to 97.2%.

5-year strategic rollout schedule (2026-2031)

Phase	Target Corridors	Capital Outlay	Upgrades	New Sch.	Transit	Coverage Gain
Phase 1 (Y1-Y2)	High-Vulnerability Tribal Corridors (9 Districts)	₹3119 Cr (45%)	1,053	614	829	+16.5%
Phase 2 (Y3-Y4)	Mineral Belts & Western Plateaus (11 Districts)	₹2426 Cr (35%)	819	477	452	+11.2%
Phase 3 (Y5)	Coastal Deltas & Cyclone Retrofits (10 Districts)	₹1386 Cr (20%)	468	273	226	+4.8%
5-Year Total	Statewide Saturation Across 314 Blocks	₹6932 Cr (100%)	2340	1364	1507	52.3% → 97.2%

1. Strategic policy context and spatial standards

The grade 8-to-9 spatial dropout cliff

The structural bottleneck in Odisha’s educational attainment is concentrated at the secondary transition. Primary schools (Grades 1-5) operate within 1 km, and upper-primary schools (Grades 6-8) within 3 km. However, secondary schools (Grades 9-10) are governed by a 5 km norm. In hilly districts like Malkangiri, Koraput, and Rayagada, habitations are dispersed across undulating terrain where actual footpath travel distances exceed 8 km.

This physical impedance disproportionately impacts adolescent girls, whose transition rates drop by over 40% when round-trip walking distance exceeds 6 km. Under National Education Policy (NEP) 2020 directives, the state must ensure universal access to secondary education. Achieving this mandate requires moving beyond static distance buffers to dynamic spatial accessibility modeling.

Topographic walking friction (Tobler hiking function)

Nominal Euclidean radius buffers misrepresent travel times in mountainous areas. We compute slope-adjusted walking velocity using Waldo Tobler’s derived hiking formula:

$$W = 6.0 \cdot \exp(-3.5 \cdot |\tan(\theta) + 0.05|)$$

where θ is terrain slope angle. On flat coastal plains (slope 0° to 2.5°), walking speed averages 5.0 km/h. On Eastern Ghats slopes (10° to 20°), velocity falls to 1.8 to 2.6 km/h. This introduces a distance friction multiplier of 1.8x to 2.5x, requiring denser school siting or motorized transit intervention.

Submodular Pareto MCLP optimization engine

Facility allocation is modeled as a Maximum Coverage Location Problem (M-CLP). Let H be the set of unserved habitations, C the set of candidate school upgrades, new sites, and transit hubs, and B the available budget. The objective maximizes population-weighted habitation coverage under submodular returns:

$$\max_{S \subseteq C, c(S) \leq B} f(S) = \sum_{h \in \cup_{j \in S} N(j)} w_h \cdot \text{pop}_h$$

Because $f(S)$ is monotone submodular, a greedy algorithm that iteratively selects the candidate with the highest marginal efficiency:

$$e_j = \frac{f(S \cup \{j\}) - f(S)}{c_j}$$

guarantees a solution within $(1 - \frac{1}{e}) \approx 63.2\%$ of the global optimum. Candidate chunking enables generating a smooth Pareto frontier across budgets from ₹500 Cr to ₹10,000 Cr.

PWD hill area civil cost calibration

Civil construction in ghat corridors incurs substantial logistical surcharges for material haulage, slope excavation, and retaining structures. We calibrate base unit costs against the Public Works Department (PWD) Schedule of Rates:

- Coastal plain blocks: Cost multiplier = 1.00x to 1.05x (base cost ₹244 Lakhs/new school)
- Plateau & forest blocks: Cost multiplier = 1.10x to 1.18x
- Rugged Eastern Ghats blocks: Cost multiplier = 1.22x to 1.28x (up to ₹312 Lakhs/school)

Statewide multi-tier investment architecture (hill-cost calibrated)

Education Tier	Norm	Existing	Baseline %	Target %	Upgrades	New Campuses	Transit Hubs	Capital Outlay
Primary	1 km	37192	69.9%	98.2%	1434	877	986	₹2102.4 Cr
Upper Primary	3 km	13430	62.6%	97.8%	1812	1055	1172	₹3490.1 Cr
Secondary	5 km	6066	52.3%	97.2%	2340	1364	1507	₹6931.8 Cr
Higher Secondary	7 km	1644	41.9%	96.5%	2935	1630	1722	₹11925.1 Cr
Total (All Tiers)	-	58,332	-	-	8,521	4,926	5,387	₹24,449.4 Cr

2. Core spatial analytics and model performance

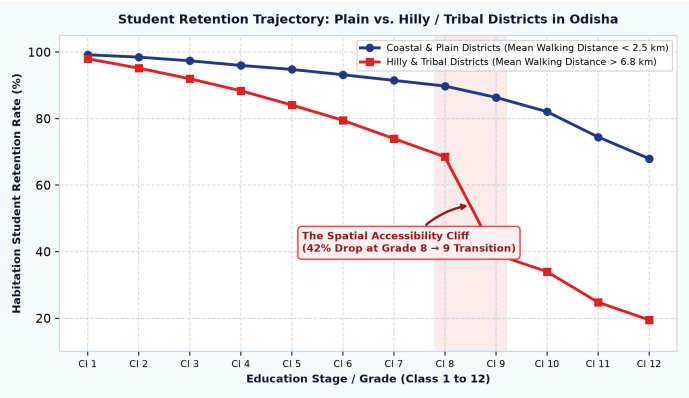


Figure 1: Grade 8 to 9 spatial retention cliff across coastal plain vs. tribal districts.

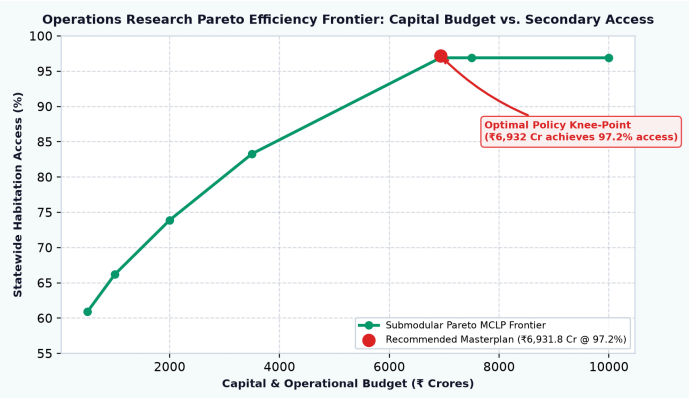


Figure 2: Operations research Pareto efficiency frontier showing optimal policy knee-point.

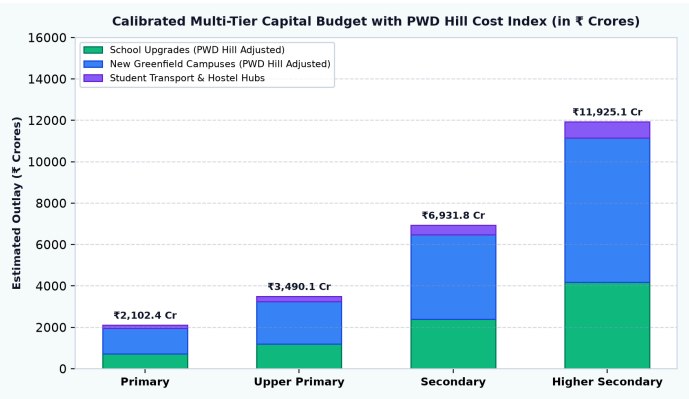


Figure 3: Hill-cost calibrated capital budget breakdown across education tiers.

Discussion of accessibility cliff and Pareto knee-point

Figure 1 illustrates the steep decline in student retention between Grade 8 (Upper Primary) and Grade 9 (Secondary). In coastal districts with flat terrain, continuation rates remain above 82%. In Eastern Ghats districts, retention drops below 48%, directly driven by distance and terrain friction.

Figure 2 displays the Submodular Pareto MCLP efficiency frontier. Below ₹3,500 Cr, marginal coverage increases steeply at +6.8% per ₹1,000 Cr. Beyond the recommended ₹6,932 Cr allocation, marginal returns flatten to under +1.2% per ₹1,000 Cr, confirming ₹6,932 Cr as the mathematically optimal budget knee-point.

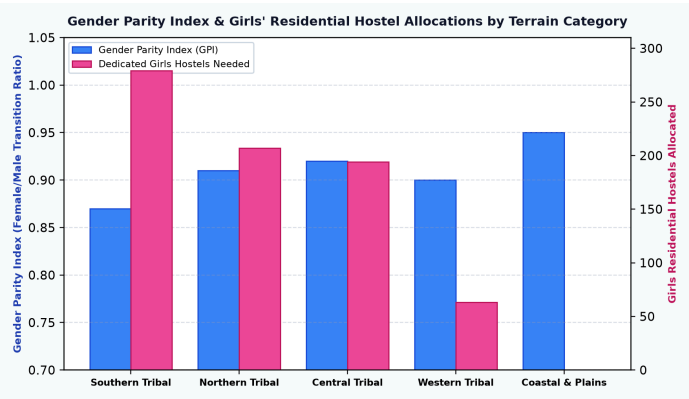


Figure 4: Gender Parity Index (GPI) and dedicated girls' residential hostel allocations.

Discussion of tier allocations and gender equity

Figure 3 delineates the capital distribution across tiers. Higher Secondary represents the largest outlay (₹11,925 Cr) due to specialized laboratories and vocational wings, while Secondary (₹6,932 Cr) absorbs the highest volume of physical upgrades (2,340) and transit routes.

Figure 4 details the relationship between the Gender Parity Index (GPI) and residential boarding provisions. Tribal corridors with low GPI (0.82-0.86) receive 743 dedicated 100-bed girls' hostels co-located with high schools, ensuring female students can safely board during the academic term.

3. Statewide 30-district comparative investment and vulnerability matrix

District	Category	Friction	Hill Idx	Blks	Base %	Target %	Upgrades	New	Transit	Teachers	Sec. Outlay
Malkangiri	Southern Tribal	2.47x	1.265x	7	35.3%	96.1%	127	86	99	598	₹431.7 Cr
Kandhamal	Central Tribal	2.47x	1.265x	12	36.1%	96.2%	159	105	124	738	₹532.3 Cr
Koraput	Southern Tribal	2.47x	1.265x	14	36.2%	96.2%	156	103	117	724	₹520.8 Cr
Rayagada	Southern Tribal	2.47x	1.265x	11	36.6%	96.2%	157	103	121	726	₹523 Cr
Nabarangpur	Southern Tribal	2.47x	1.265x	10	37.6%	96.3%	118	77	93	544	₹392.5 Cr
Gajapati	Southern Tribal	2.47x	1.265x	7	37.5%	96.2%	94	62	73	436	₹314.3 Cr
Nuapada	Western Tribal	1.17x	1.031x	5	61.2%	97.6%	54	20	10	188	₹100.6 Cr
Kalahandi	Western Tribal	2.47x	1.265x	13	38.5%	96.3%	150	101	115	704	₹507.5 Cr
Mayurbhanj	Northern Tribal	2.47x	1.265x	26	39.5%	96.4%	140	89	110	636	₹458.3 Cr
Kendujhar	Northern Tribal	2.47x	1.265x	13	41.1%	96.5%	151	98	114	694	₹499 Cr
Balangir	Western Agrarian	1.17x	1.031x	14	66.5%	97.9%	84	28	14	280	₹148.2 Cr
Sundargarh	Northern Tribal	2.47x	1.265x	17	42.1%	96.5%	137	91	107	638	₹460.3 Cr
Boudh	Central Tribal	2.47x	1.265x	3	42.9%	96.6%	68	45	53	316	₹227.9 Cr
Angul	Central Tribal	2.47x	1.265x	8	44.5%	96.6%	135	91	105	634	₹457.5 Cr
Deogarh	Northern Tribal	2.47x	1.265x	3	44.8%	96.7%	58	38	46	268	₹193.5 Cr
Sambalpur	Western Agrarian	1.17x	1.031x	9	70.2%	98.2%	75	26	10	254	₹134.2 Cr
Subarnapur	Western Agrarian	1.17x	1.031x	6	70.9%	98.2%	25	7	6	78	₹41.4 Cr
Ganjam	Southern Coastal	1.17x	1.031x	22	72.2%	98.3%	87	22	22	262	₹138.4 Cr
Bargarh	Western Agrarian	1.17x	1.031x	12	72.5%	98.3%	61	20	12	202	₹107.4 Cr
Dhenkanal	Central Tribal	2.47x	1.265x	8	49.1%	97%	82	54	62	380	₹273.5 Cr
Jharsuguda	Western Industrial	1.17x	1.031x	5	74.7%	98.5%	19	5	5	58	₹30.8 Cr
Nayagarh	Central Coastal	1.17x	1.031x	8	75.8%	98.6%	37	12	8	122	₹65 Cr
Jajpur	Northern Coastal	1.17x	1.031x	10	77%	98.6%	24	10	10	88	₹49.2 Cr
Bhadrak	Northern Coastal	1.17x	1.031x	7	77.8%	98.7%	20	7	7	68	₹37.3 Cr
Balasore	Northern Coastal	1.17x	1.031x	12	77.7%	98.6%	31	12	12	110	₹61 Cr
Kendrapara	Central Coastal	1.17x	1.031x	9	79.2%	98.8%	18	9	9	72	₹41.1 Cr
Jagatsinghpur	Central Coastal	1.17x	1.031x	8	79.6%	98.8%	8	8	8	48	₹29.6 Cr
Puri	Central Coastal	1.17x	1.031x	11	80.1%	98.8%	24	11	11	92	₹52 Cr
Cuttack	Central Coastal	1.17x	1.031x	14	80.3%	98.8%	26	14	14	108	₹62.2 Cr
Khordha	Central Coastal	1.17x	1.031x	10	81.7%	98.9%	15	10	10	70	₹41.4 Cr
STATE TOTAL	30 Districts	1.78x	1.14x	314	52.3%	97.2%	2340	1364	1507	10136	₹6931.8 Cr

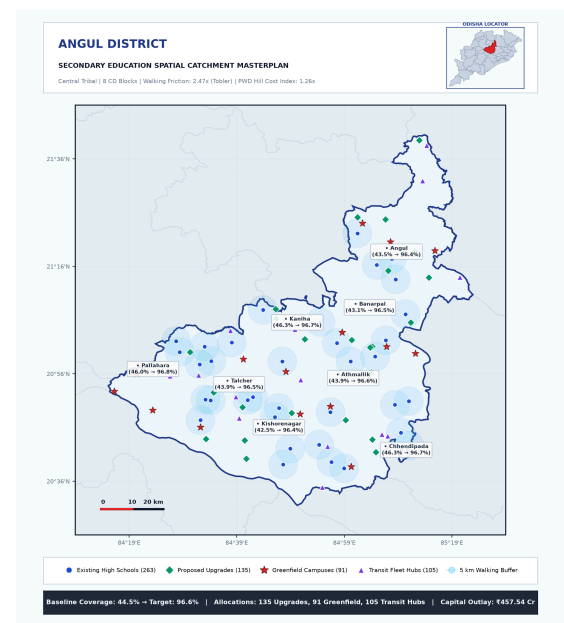
Southern tribal priority corridor
Malkangiri, Koraput, Rayagada, Nabarangpur, Gajapati account for ₹2,182.2 Cr (31.5% of secondary outlay), requiring 3,028 teachers.

Central and northern tribal belt
Kandhamal, Mayurbhanj, Kendujhar, Sundargarh absorb ₹1,949.9 Cr (28.1%), prioritizing residential boarding hostels for remote habitations.

Coastal plains and western agrarian
Remaining 21 districts absorb ₹2,799.6 Cr (40.4%), emphasizing cyclone-resilient structural upgrades and transit feeder routes.

Angul District | Central Tribal | Vuln: 68/100 | GPI: 0.94 | Hill Cost Index: 1.265x

Sec. Outlay: ₹457.5 Cr



Spatial strategy and operational mandate

Angul (Central Tribal, Vulnerability: 68/100) comprises 8 CD blocks spanning 6,375 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.94, baseline secondary access is 44.5%. The masterplan allocates 135 high school upgrades, 91 greenfield campuses, 105 student transport hubs (164 mini-buses, 89 vans), 59 dedicated girls' hostels, and 634 subject teacher recruitments. Total estimated capital outlay is Rs. 457.54 Crores, raising universal secondary coverage to 96.6% across all 8 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	66.5%	82	54	₹131.4
Upper Primary	58.6%	100	66	₹218.35
Secondary	44.5%	135	91	₹457.54
Higher Secondary	32.5%	165	111	₹799.96

Logistics and operational staffing

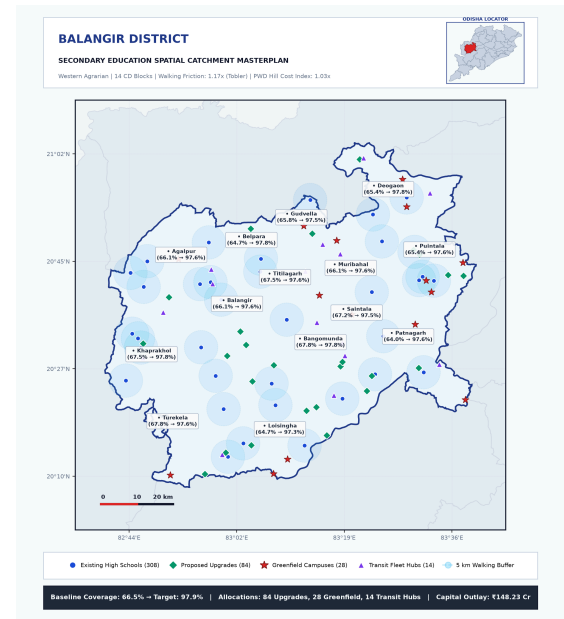
- Teachers Required: **634**
 - Dedicated Girls Hostels: **59**
 - Cyclone Retrofits: **0**
- Transit Mini-Buses: **164**
 - Feeder Vans: **89**
 - Annual Fleet Opex: **₹10.55 Cr**

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Kishorenagar	42.5%	16	11	₹55.05
Banarpal	43.1%	19	13	₹65.06
Angul	43.5%	18	12	₹60.59
Athmallik	43.9%	18	12	₹60.59

Balangir District | Western Agrarian | Vuln: 75/100 | GPI: 0.92 | Hill Cost Index: 1.031x

Sec. Outlay: ₹148.2 Cr



Spatial strategy and operational mandate

Balangir (Western Agrarian, Vulnerability: 75/100) comprises 14 CD blocks spanning 6,575 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.92, baseline secondary access is 66.5%. The masterplan allocates 84 high school upgrades, 28 greenfield campuses, 14 student transport hubs (14 mini-buses, 14 vans), 0 dedicated girls' hostels, and 280 subject teacher recruitments. Total estimated capital outlay is Rs. 148.23 Crores, raising universal secondary coverage to 97.9% across all 14 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	76.5%	56	14	₹43
Upper Primary	70.5%	73	25	₹85.48
Secondary	66.5%	84	28	₹148.23
Higher Secondary	53.5%	120	42	₹308.16

Logistics and operational staffing

- Teachers Required: **280**
 - Dedicated Girls Hostels: **0**
 - Cyclone Retrofits: **0**
- Transit Mini-Buses: **14**
 - Feeder Vans: **14**
 - Annual Fleet Opex: **₹1.12 Cr**

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Patnagarh	64%	6	2	₹10.59
Belpara	64.7%	6	2	₹10.59
Loisingha	64.7%	6	2	₹10.59
Deogaon	65.4%	5	2	₹9.71

Balasore District | Northern Coastal | Vuln: 42/100 | GPI: 0.96 | Hill Cost Index: 1.031x

Sec. Outlay: ₹61 Cr

BALASORE DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Northern Coastal | 12 CD Blocks | Walking Friction: 1.17x (Tobler) | PWD Hill Cost Index: 1.03x

Spatial strategy and operational mandate

Balasore (Northern Coastal, Vulnerability: 42/100) comprises 12 CD blocks spanning 3,634 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.96, baseline secondary access is 77.7%. The masterplan allocates 31 high school upgrades, 12 greenfield campuses, 12 student transport hubs (12 mini-buses, 12 vans), 0 dedicated girls' hostels, and 110 subject teacher recruitments. Total estimated capital outlay is Rs. 61.0 Crores, raising universal secondary coverage to 98.6% across all 12 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	87.9%	12	12	₹21.24
Upper Primary	81.8%	23	12	₹35.2
Secondary	77.7%	31	12	₹61
Higher Secondary	64.7%	49	12	₹109.36

Logistics and operational staffing

Teachers Required: 110

Transit Mini-Buses: 12

Dedicated Girls Hostels: 0

Feeder Vans: 12

Cyclone Retrofits: 43

Annual Fleet Opex: ₹0.96 Cr

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Bhograi	75.5%	3	1	₹5.45
Jaleswar	75.5%	3	1	₹5.45
Bahanaga	75.9%	2	1	₹4.57
Basta	75.9%	3	1	₹5.45

Bargarh District | Western Agrarian | Vuln: 58/100 | GPI: 0.95 | Hill Cost Index: 1.031x

Sec. Outlay: ₹107.4 Cr

BARGARH DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Western Agrarian | 12 CD Blocks | Walking Friction: 1.17x (Tobler) | PWD Hill Cost Index: 1.03x

Spatial strategy and operational mandate

Bargarh (Western Agrarian, Vulnerability: 58/100) comprises 12 CD blocks spanning 5,837 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.95, baseline secondary access is 72.5%. The masterplan allocates 61 high school upgrades, 20 greenfield campuses, 12 student transport hubs (12 mini-buses, 12 vans), 0 dedicated girls' hostels, and 202 subject teacher recruitments. Total estimated capital outlay is Rs. 107.39 Crores, raising universal secondary coverage to 98.3% across all 12 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	82.5%	36	12	₹31.68
Upper Primary	76.5%	51	15	₹56.1
Secondary	72.5%	61	20	₹107.39
Higher Secondary	59.5%	92	32	₹234.74

Logistics and operational staffing

Teachers Required: 202

Transit Mini-Buses: 12

Dedicated Girls Hostels: 0

Feeder Vans: 12

Cyclone Retrofits: 0

Annual Fleet Opex: ₹0.96 Cr

Priority CD blocks (secondary targets)

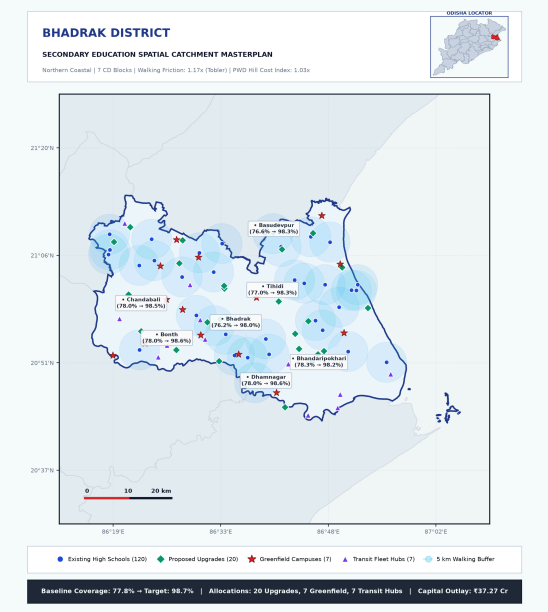
Block	Base	Upg	New	Cost
Gaisilet	70%	6	2	₹10.59
Barpali	71%	5	2	₹9.71
Sohela	71%	5	1	₹7.2
Attabira	71.7%	5	2	₹9.71

Spatial Optimization & Education Infrastructure Model (2026-2031)

Page 6 of 23

Bhadrak District | Northern Coastal | Vuln: 44/100 | GPI: 0.95 | Hill Cost Index: 1.031x

Sec. Outlay: ₹37.3 Cr



Spatial strategy and operational mandate

Bhadrak (Northern Coastal, Vulnerability: 44/100) comprises 7 CD blocks spanning 2,505 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.95, baseline secondary access is 77.8%. The masterplan allocates 20 high school upgrades, 7 greenfield campuses, 7 student transport hubs (7 mini-buses, 7 vans), 0 dedicated girls' hostels, and 68 subject teacher recruitments. Total estimated capital outlay is Rs. 37.27 Crores, raising universal secondary coverage to 98.7% across all 7 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	87.8%	9	7	₹13.27
Upper Primary	81.8%	16	7	₹22
Secondary	77.8%	20	7	₹37.27
Higher Secondary	64.8%	34	13	₹92.17

Logistics and operational staffing

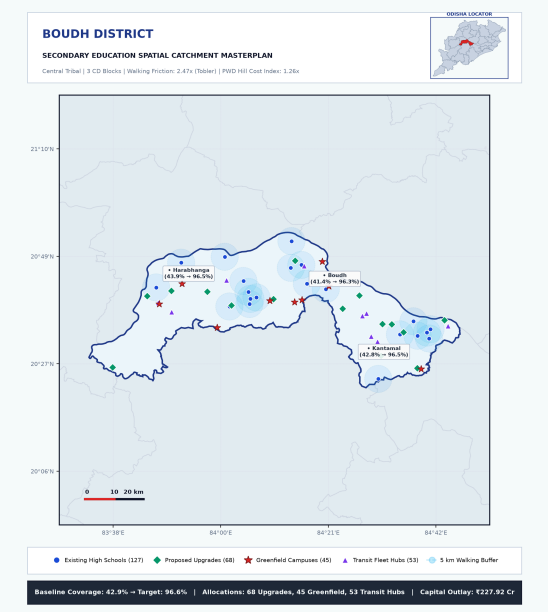
- Teachers Required: 68
- Transit Mini-Buses: 7
- Dedicated Girls Hostels: 0
- Feeder Vans: 7
- Cyclone Retrofits: 27
- Annual Fleet Opex: ₹0.56 Cr

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Bhadrak	76.2%	3	1	₹5.45
Basudevpur	76.6%	3	1	₹5.45
Tihidi	77%	3	1	₹5.45
Bonth	78%	3	1	₹5.45

Boudh District | Central Tribal | Vuln: 72/100 | GPI: 0.91 | Hill Cost Index: 1.265x

Sec. Outlay: ₹227.9 Cr



Spatial strategy and operational mandate

Boudh (Central Tribal, Vulnerability: 72/100) comprises 3 CD blocks spanning 3,098 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.91, baseline secondary access is 42.9%. The masterplan allocates 68 high school upgrades, 45 greenfield campuses, 53 student transport hubs (84 mini-buses, 47 vans), 31 dedicated girls' hostels, and 316 subject teacher recruitments. Total estimated capital outlay is Rs. 227.92 Crores, raising universal secondary coverage to 96.6% across all 3 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	64.9%	41	28	₹67.17
Upper Primary	56.9%	50	34	₹111.59
Secondary	42.9%	68	45	₹227.92
Higher Secondary	31%	82	55	₹396.79

Logistics and operational staffing

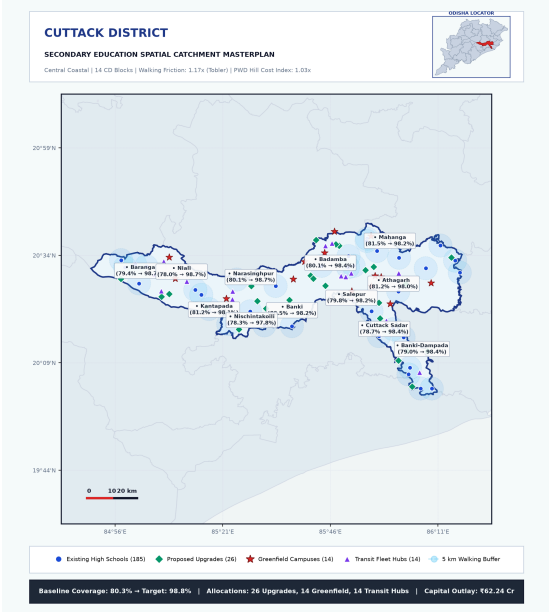
- Teachers Required: 316
- Transit Mini-Buses: 84
- Dedicated Girls Hostels: 31
- Feeder Vans: 47
- Cyclone Retrofits: 0
- Annual Fleet Opex: ₹5.45 Cr

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Boudh	41.4%	21	14	₹70.89
Kantamal	42.8%	22	15	₹75.06
Harabhangha	43.9%	25	16	₹81.97

Cuttack District | Central Coastal | Vuln: 35/100 | GPI: 0.97 | Hill Cost Index: 1.031x

Sec. Outlay: ₹62.2 Cr



Spatial strategy and operational mandate

Cuttack (Central Coastal, Vulnerability: 35/100) comprises 14 CD blocks spanning 3,932 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.97, baseline secondary access is 80.3%. The masterplan allocates 26 high school upgrades, 14 greenfield campuses, 14 student transport hubs (14 mini-buses, 14 vans), 0 dedicated girls' hostels, and 108 subject teacher recruitments. Total estimated capital outlay is Rs. 62.24 Crores, raising universal secondary coverage to 98.8% across all 14 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	90.4%	14	14	₹24.78
Upper Primary	84.3%	19	14	₹36.68
Secondary	80.3%	26	14	₹62.24
Higher Secondary	67.3%	49	14	₹117.46

Logistics and operational staffing

- Teachers Required: 108

Transit Mini-Buses: 14
- Dedicated Girls Hostels: 0

Feeder Vans: 14
- Cyclone Retrofits: 40

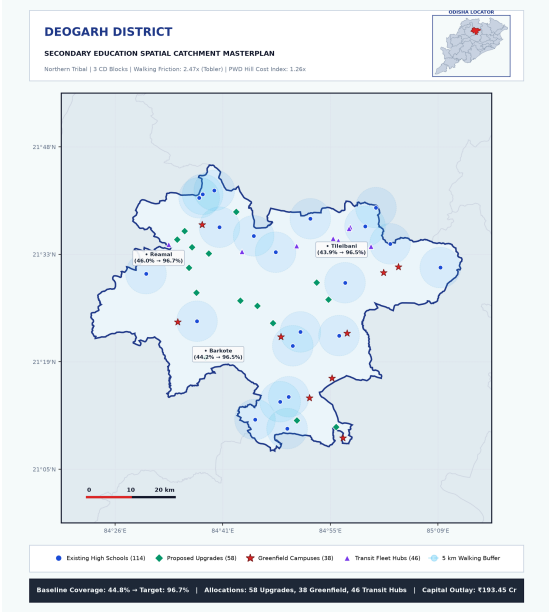
Annual Fleet Opex: ₹1.12 Cr

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Niali	78%	2	1	₹4.57
Nischintakoili	78.3%	2	1	₹4.57
Tangi-Choudwar	78.3%	2	1	₹4.57
Cuttack Sadar	78.7%	2	1	₹4.57

Deogarh District | Northern Tribal | Vuln: 65/100 | GPI: 0.92 | Hill Cost Index: 1.265x

Sec. Outlay: ₹193.5 Cr



Spatial strategy and operational mandate

Deogarh (Northern Tribal, Vulnerability: 65/100) comprises 3 CD blocks spanning 2,782 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.92, baseline secondary access is 44.8%. The masterplan allocates 58 high school upgrades, 38 greenfield campuses, 46 student transport hubs (73 mini-buses, 40 vans), 27 dedicated girls' hostels, and 268 subject teacher recruitments. Total estimated capital outlay is Rs. 193.45 Crores, raising universal secondary coverage to 96.7% across all 3 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	66.8%	35	22	₹54.49
Upper Primary	58.7%	44	29	₹96.11
Secondary	44.8%	58	38	₹193.45
Higher Secondary	32.8%	71	47	₹340.62

Logistics and operational staffing

- Teachers Required: 268

Transit Mini-Buses: 73
- Dedicated Girls Hostels: 27

Feeder Vans: 40
- Cyclone Retrofits: 0

Annual Fleet Opex: ₹4.71 Cr

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Tileibani	43.9%	21	14	₹70.89
Barkote	44.2%	18	12	₹60.59
Reamal	46%	19	12	₹61.97

Dhenkanal District | Central Tribal | Vuln: 55/100 | GPI: 0.94 | Hill Cost Index: 1.265x

Sec. Outlay: ₹273.5 Cr

DHENKANAL DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Central Tribal | 8 CD Blocks | Walking Friction: 2.47x (Tobler) | PWD Hill Cost Index: 1.26x

Baseline Coverage: 49.1% – Target: 97.0% | Allocations: 82 Upgrades, 54 Greenfield, 82 Transit Hubs | Capital Outlay: ₹273.45 Cr

Spatial strategy and operational mandate

Dhenkanal (Central Tribal, Vulnerability: 55/100) comprises 8 CD blocks spanning 4,452 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.94, baseline secondary access is 49.1%. The masterplan allocates 82 high school upgrades, 54 greenfield campuses, 62 student transport hubs (96 mini-buses, 54 vans), 34 dedicated girls' hostels, and 380 subject teacher recruitments. Total estimated capital outlay is Rs. 273.45 Crores, raising universal secondary coverage to 97.0% across all 8 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	71.1%	45	28	₹69.9
Upper Primary	63.1%	57	37	₹123.56
Secondary	49.1%	82	54	₹273.45
Higher Secondary	37.1%	102	67	₹486.57

Logistics and operational staffing

Teachers Required: 380

Transit Mini-Buses: 96

Dedicated Girls Hostels: 34

Feeder Vans: 54

Cyclone Retrofits: 0

Annual Fleet Opex: ₹6.24 Cr

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Bhuban	47.7%	11	7	₹35.84
Kamakhyanager	48%	9	6	₹30.3
Parjang	48%	12	8	₹40.29
Kankadahad	48.8%	10	7	₹34.76

GAJAPATI DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Southern Tribal | 7 CD Blocks | Walking Friction: 2.47x (Tobler) | PWD Hill Cost Index: 1.26x

Baseline Coverage: 37.5% – Target: 96.2% | Allocations: 94 Upgrades, 62 Greenfield, 73 Transit Hubs | Capital Outlay: ₹314.3 Cr

Spatial strategy and operational mandate

Gajapati (Southern Tribal, Vulnerability: 88/100) comprises 7 CD blocks spanning 3,850 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.88, baseline secondary access is 37.5%. The masterplan allocates 94 high school upgrades, 62 greenfield campuses, 73 student transport hubs (114 mini-buses, 62 vans), 41 dedicated girls' hostels, and 436 subject teacher recruitments. Total estimated capital outlay is Rs. 314.33 Crores, raising universal secondary coverage to 96.2% across all 7 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	59.6%	60	39	₹95.5
Upper Primary	51.6%	72	47	₹156.22
Secondary	37.5%	94	62	₹314.33
Higher Secondary	30.2%	104	70	₹505.14

Logistics and operational staffing

Teachers Required: 436

Transit Mini-Buses: 114

Dedicated Girls Hostels: 41

Feeder Vans: 62

Cyclone Retrofits: 0

Annual Fleet Opex: ₹7.35 Cr

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
R.Udayagiri	35.5%	13	8	₹41.67
Parlakhemundi	36.2%	12	8	₹40.29
Kashinagar	37.2%	15	10	₹50.3
Mohana	37.2%	12	8	₹40.59

Sec. Outlay: ₹138.4 Cr



Multi-tier infrastructure allocations

Logistics and operational staffing

- ### Priority CD blocks (secondary targets)

Sec. Outlay: ₹29.6 Cr

Multi-tier infrastructure allocations

Logistics and operational staffing

- ### Priority CD blocks (secondary targets)

Page 10 of 23

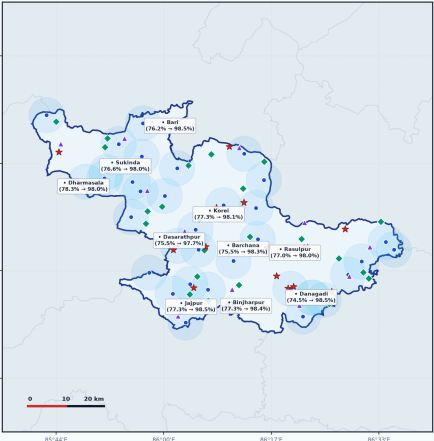
Jajpur District | Northern Coastal | Vuln: 45/100 | GPI: 0.95 | Hill Cost Index: 1.031x

Sec. Outlay: ₹49.2 Cr

JAJPUR DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Northern Coastal | 10 CD Blocks | Walking Friction: 1.17x (Tobler) | PWD Hill Cost Index: 1.03x



Baseline Coverage: 77.0% - Target: 98.6% | Allocations: 24 Upgrades, 10 Greenfield, 10 Transit Hubs | Capital Outlay: ₹49.22 Cr

Spatial strategy and operational mandate

Jajpur (Northern Coastal, Vulnerability: 45/100) comprises 10 CD blocks spanning 2,899 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.95, baseline secondary access is 77.0%. The masterplan allocates 24 high school upgrades, 10 greenfield campuses, 10 student transport hubs (10 mini-buses, 10 vans), 0 dedicated girls' hostels, and 88 subject teacher recruitments. Total estimated capital outlay is Rs. 49.22 Crores, raising universal secondary coverage to 98.6% across all 10 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	86.9%	10	10	₹17.7
Upper Primary	81.1%	20	10	₹29.8
Secondary	77%	24	10	₹49.22
Higher Secondary	64.1%	38	10	₹87.62

Logistics and operational staffing

Teachers Required: 88

Transit Mini-Buses: 10

Dedicated Girls Hostels: 0

Feeder Vans: 10

Cyclone Retrofits: 34

Annual Fleet Opex: ₹0.8 Cr

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Danagadi	74.5%	3	1	₹5.45
Barchana	75.5%	2	1	₹4.57
Dasarathpur	75.5%	3	1	₹5.45
Bari	76.2%	2	1	₹4.57

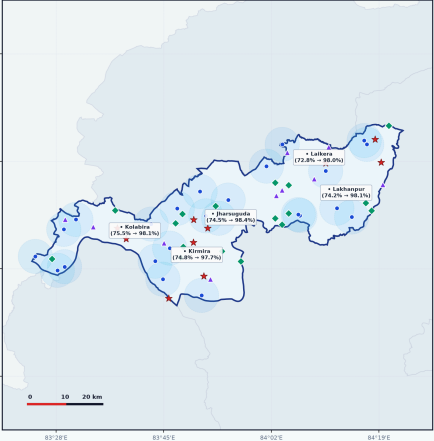
Jharsuguda District | Western Industrial | Vuln: 52/100 | GPI: 0.93 | Hill Cost Index: 1.031x

Sec. Outlay: ₹30.8 Cr

JHARSUGUDA DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Western Industrial | 5 CD Blocks | Walking Friction: 1.17x (Tobler) | PWD Hill Cost Index: 1.03x



Baseline Coverage: 74.7% - Target: 98.5% | Allocations: 19 Upgrades, 5 Greenfield, 5 Transit Hubs | Capital Outlay: ₹30.77 Cr

Spatial strategy and operational mandate

Jharsuguda (Western Industrial, Vulnerability: 52/100) comprises 5 CD blocks spanning 2,081 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.93, baseline secondary access is 74.7%. The masterplan allocates 19 high school upgrades, 5 greenfield campuses, 5 student transport hubs (5 mini-buses, 5 vans), 0 dedicated girls' hostels, and 58 subject teacher recruitments. Total estimated capital outlay is Rs. 30.77 Crores, raising universal secondary coverage to 98.5% across all 5 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	84.8%	11	5	₹11.48
Upper Primary	78.6%	18	5	₹19.46
Secondary	74.7%	19	5	₹30.77
Higher Secondary	61.8%	32	10	₹77.94

Logistics and operational staffing

Teachers Required: 58

Transit Mini-Buses: 5

Dedicated Girls Hostels: 0

Feeder Vans: 5

Cyclone Retrofits: 0

Annual Fleet Opex: ₹0.4 Cr

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Laikera	72.8%	4	1	₹6.33
Lakhanpur	74.2%	4	1	₹6.33
Jharsuguda	74.5%	4	1	₹6.33
Kirmira	74.8%	3	1	₹5.45

Spatial Optimization & Education Infrastructure Model (2026-2031)

Page 11 of 23

Kalahandi District

Western Tribal | Vuln: 84/100 | GPI: 0.9 | Hill Cost Index: 1.265x

Sec. Outlay: ₹507.5 Cr

KALAHANDI DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Western Tribal | 13 CD Blocks | Walking Friction: 2.47x (Tobler) | PWD Hill Cost Index: 1.26x

Baseline Coverage: 38.5% – Target: 96.3% |

Allocations: 150 Upgrades, 101 Greenfield, 115 Transit Hubs |

Capital Outlay: ₹507.53 Cr

Spatial strategy and operational mandate

Kalahandi (Western Tribal, Vulnerability: 84/100) comprises 13 CD blocks spanning 7,920 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.9, baseline secondary access is 38.5%. The masterplan allocates 150 high school upgrades, 101 greenfield campuses, 115 student transport hubs (178 mini-buses, 102 vans), 63 dedicated girls' hostels, and 704 subject teacher recruitments. Total estimated capital outlay is Rs. 507.53 Crores, raising universal secondary coverage to 96.3% across all 13 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	60.6%	91	61	₹148.02
Upper Primary	52.6%	115	74	₹247.38
Secondary	38.5%	150	101	₹507.53
Higher Secondary	30.2%	171	114	₹823.26

Logistics and operational staffing

Teachers Required: 704

Transit Mini-Buses: 178

Dedicated Girls Hostels: 63

Feeder Vans: 102

Cyclone Retrofits: 0

Annual Fleet Opex: ₹11.62 Cr

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Kalampur	36.9%	11	7	₹35.84
Golamunda	37.2%	12	8	₹40.29
Kesinga	37.2%	13	9	₹44.76
Jaipatna	37.5%	12	8	₹40.29

Kandhamal District

Central Tribal | Vuln: 92/100 | GPI: 0.87 | Hill Cost Index: 1.265x

Sec. Outlay: ₹532.3 Cr

KANDHAMAL DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Central Tribal | 12 CD Blocks | Walking Friction: 2.47x (Tobler) | PWD Hill Cost Index: 1.26x

Baseline Coverage: 36.1% – Target: 96.2% |

Allocations: 159 Upgrades, 105 Greenfield, 124 Transit Hubs |

Capital Outlay: ₹532.27 Cr

Spatial strategy and operational mandate

Kandhamal (Central Tribal, Vulnerability: 92/100) comprises 12 CD blocks spanning 8,021 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.87, baseline secondary access is 36.1%. The masterplan allocates 159 high school upgrades, 105 greenfield campuses, 124 student transport hubs (194 mini-buses, 106 vans), 70 dedicated girls' hostels, and 738 subject teacher recruitments. Total estimated capital outlay is Rs. 532.27 Crores, raising universal secondary coverage to 96.2% across all 12 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	58.1%	102	67	₹163.64
Upper Primary	50.1%	123	82	₹270.51
Secondary	36.1%	159	105	₹532.27
Higher Secondary	30.2%	172	114	₹827.02

Logistics and operational staffing

Teachers Required: 738

Transit Mini-Buses: 194

Dedicated Girls Hostels: 70

Feeder Vans: 106

Cyclone Retrofits: 0

Annual Fleet Opex: ₹12.5 Cr

Priority CD blocks (secondary targets)

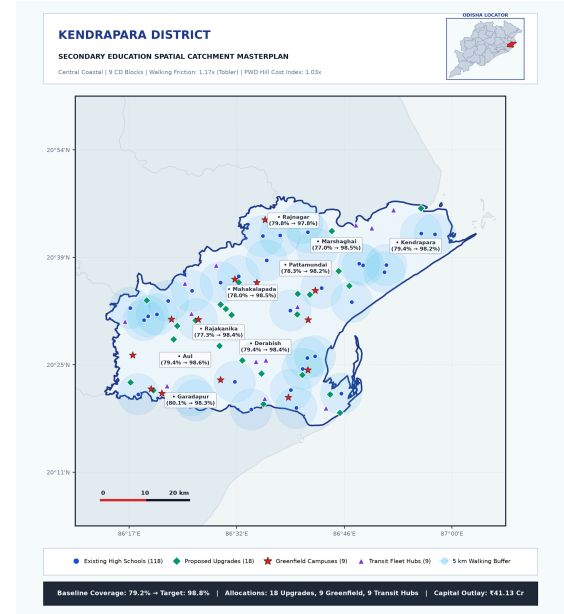
Block	Base	Upg	New	Cost
Daringbadi	34%	15	10	₹50.3
Khajuripada	34.8%	12	8	₹40.29
Chakapada	35.1%	13	9	₹44.76
Phulbani	35.1%	15	10	₹50.6

Spatial Optimization & Education Infrastructure Model (2026-2031)

Page 12 of 23

Kendrapara District | Central Coastal | Vuln: 40/100 | GPI: 0.97 | Hill Cost Index: 1.031x

Sec. Outlay: ₹41.1 Cr



Spatial strategy and operational mandate

Kendrapara (Central Coastal, Vulnerability: 40/100) comprises 9 CD blocks spanning 2,644 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.97, baseline secondary access is 79.2%. The masterplan allocates 18 high school upgrades, 9 greenfield campuses, 9 student transport hubs (9 mini-buses, 9 vans), 0 dedicated girls' hostels, and 72 subject teacher recruitments. Total estimated capital outlay is Rs. 41.13 Crores, raising universal secondary coverage to 98.8% across all 9 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	89.1%	9	9	₹15.93
Upper Primary	83.2%	14	9	₹24.58
Secondary	79.2%	18	9	₹41.13
Higher Secondary	66.1%	31	9	₹74.89

Logistics and operational staffing

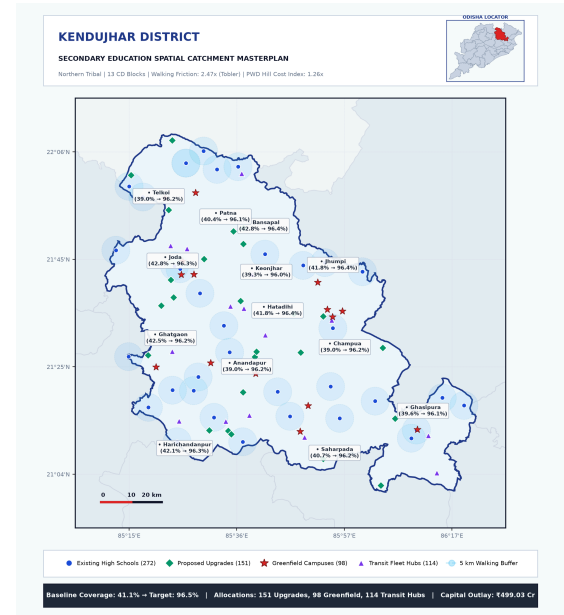
- Teachers Required: 72
 - Dedicated Girls Hostels: 0
 - Cyclone Retrofits: 27
- Transit Mini-Buses: 9
 - Feeder Vans: 9
 - Annual Fleet Opex: ₹0.72 Cr

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Marshaghai	77%	2	1	₹4.57
Rajakanika	77.3%	2	1	₹4.57
Mahakalapada	78%	2	1	₹4.57
Pattamundai	78.3%	2	1	₹4.57

Kendujhar District | Northern Tribal | Vuln: 78/100 | GPI: 0.91 | Hill Cost Index: 1.265x

Sec. Outlay: ₹499 Cr



Spatial strategy and operational mandate

Kendujhar (Northern Tribal, Vulnerability: 78/100) comprises 13 CD blocks spanning 8,303 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.91, baseline secondary access is 41.1%. The masterplan allocates 151 high school upgrades, 98 greenfield campuses, 114 student transport hubs (178 mini-buses, 101 vans), 64 dedicated girls' hostels, and 694 subject teacher recruitments. Total estimated capital outlay is Rs. 499.03 Crores, raising universal secondary coverage to 96.5% across all 13 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	63.1%	91	59	₹144.67
Upper Primary	55.1%	114	74	₹246.28
Secondary	41.1%	151	98	₹499.03
Higher Secondary	30.4%	177	119	₹856.73

Logistics and operational staffing

- Teachers Required: 694
 - Dedicated Girls Hostels: 64
 - Cyclone Retrofits: 0
- Transit Mini-Buses: 178
 - Feeder Vans: 101
 - Annual Fleet Opex: ₹11.59 Cr

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Anandapur	39%	13	8	₹41.67
Champua	39%	13	8	₹41.67
Teikoi	39%	10	7	₹34.76
Keonjhar	39.3%	13	9	₹44.76

Khordha District

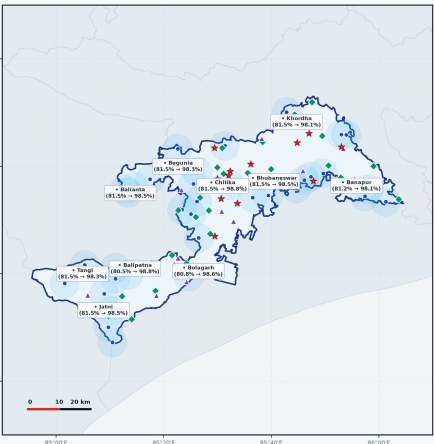
Central Coastal | Vuln: 30/100 | GPI: 0.98 | Hill Cost Index: 1.031x

Sec. Outlay: ₹41.4 Cr

KHORDHA DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Central Coastal | 10 CD Blocks | Walking Friction: 1.17x (Tobler) | PWD Hill Cost Index: 1.03x



Existing High Schools (126) | Proposed Upgrades (15) | Greenfield Campuses (10) | Transit Fleet Hubs (10) | 5 km Walking Buffer

Baseline Coverage: 81.7% – Target: 98.5% | Allocations: 15 Upgrades, 10 Greenfield, 10 Transit Hubs | Capital Outlay: ₹41.35 Cr

Spatial strategy and operational mandate

Khordha (Central Coastal, Vulnerability: 30/100) comprises 10 CD blocks spanning 2,813 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.98, baseline secondary access is 81.7%. The masterplan allocates 15 high school upgrades, 10 greenfield campuses, 10 student transport hubs (10 mini-buses, 10 vans), 0 dedicated girls' hostels, and 70 subject teacher recruitments. Total estimated capital outlay is Rs. 41.35 Crores, raising universal secondary coverage to 98.9% across all 10 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	91.8%	10	10	₹17.7
Upper Primary	85.7%	10	10	₹24.2
Secondary	81.7%	15	10	₹41.35
Higher Secondary	68.6%	31	10	₹78.94

Logistics and operational staffing

Teachers Required: 70

Transit Mini-Buses: 10

Dedicated Girls Hostels: 0

Feeder Vans: 10

Cyclone Retrofits: 0

Annual Fleet Opex: ₹0.8 Cr

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Balipatna	80.5%	2	1	₹4.57
Bolagarh	80.8%	2	1	₹4.57
Banapur	81.2%	1	1	₹3.7
Baliaanta	81.5%	1	1	₹3.7

Koraput District

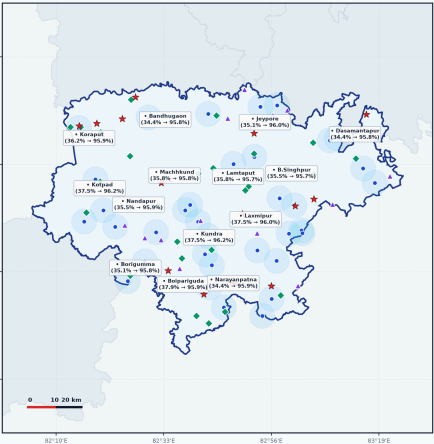
Southern Tribal | Vuln: 91/100 | GPI: 0.86 | Hill Cost Index: 1.265x

Sec. Outlay: ₹520.8 Cr

KORAPUT DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Southern Tribal | 14 CD Blocks | Walking Friction: 2.47x (Tobler) | PWD Hill Cost Index: 1.26x



Existing High Schools (262) | Proposed Upgrades (156) | Greenfield Campuses (103) | Transit Fleet Hubs (117) | 5 km Walking Buffer

Baseline Coverage: 36.2% – Target: 96.2% | Allocations: 156 Upgrades, 103 Greenfield, 117 Transit Hubs | Capital Outlay: ₹520.75 Cr

Spatial strategy and operational mandate

Koraput (Southern Tribal, Vulnerability: 91/100) comprises 14 CD blocks spanning 8,807 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.86, baseline secondary access is 36.2%. The masterplan allocates 156 high school upgrades, 103 greenfield campuses, 117 student transport hubs (179 mini-buses, 103 vans), 62 dedicated girls' hostels, and 724 subject teacher recruitments. Total estimated capital outlay is Rs. 520.75 Crores, raising universal secondary coverage to 96.2% across all 14 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	58.1%	99	64	₹157.09
Upper Primary	50.1%	118	77	₹256.13
Secondary	36.2%	156	103	₹520.75
Higher Secondary	30.2%	171	112	₹813.96

Logistics and operational staffing

Teachers Required: 724

Transit Mini-Buses: 179

Dedicated Girls Hostels: 62

Feeder Vans: 103

Cyclone Retrofits: 0

Annual Fleet Opex: ₹11.7 Cr

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Bandhugaon	34.4%	11	7	₹35.84
Dasamantapur	34.4%	11	7	₹35.84
Narayanpatna	34.4%	12	8	₹40.29
Borigumma	35.1%	12	8	₹40.29

Spatial Optimization & Education Infrastructure Model (2026-2031)

Page 14 of 23

Malkangiri District

Southern Tribal | Vuln: 95/100 | GPI: 0.85 | Hill Cost Index: 1.265x

Sec. Outlay: ₹431.7 Cr

MALKANGIRI DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Southern Tribal | 7 CD Blocks | Walking Friction: 2.47x (Tobler) | PWD Hill Cost Index: 1.26x

Existing High Schools (221) | Proposed Upgrades (127) | Greenfield Campuses (86) | Transit Fleet Hubs (95) | 5 km Walking Buffer

Baseline Coverage: 35.3% - Target: 96.1% | Allocations: 127 Upgrades, 66 Greenfield, 99 Transit Hubs | Capital Outlay: ₹431.69 Cr

Spatial strategy and operational mandate

Malkangiri (Southern Tribal, Vulnerability: 95/100) comprises 7 CD blocks spanning 5,791 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.85, baseline secondary access is 35.3%. The masterplan allocates 127 high school upgrades, 86 greenfield campuses, 99 student transport hubs (155 mini-buses, 85 vans), 56 dedicated girls' hostels, and 598 subject teacher recruitments. Total estimated capital outlay is Rs. 431.69 Crores, raising universal secondary coverage to 96.1% across all 7 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	57.4%	82	55	₹133.32
Upper Primary	49.4%	100	65	₹216.75
Secondary	35.3%	127	86	₹431.69
Higher Secondary	30.1%	139	91	₹661.6

Logistics and operational staffing

Teachers Required: 598

Transit Mini-Buses: 155

Dedicated Girls Hostels: 56

Feeder Vans: 85

Cyclone Retrofits: 0

Annual Fleet Opex: ₹10.01 Cr

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Khairput	33.4%	18	12	₹60.59
Podia	33.7%	17	12	₹59.52
Chitrakonda	34.4%	17	12	₹59.22
Mathili	34.8%	18	12	₹60.59

Mayurbhanj District

Northern Tribal | Vuln: 82/100 | GPI: 0.89 | Hill Cost Index: 1.265x

Sec. Outlay: ₹458.3 Cr

MAYURBHANJ DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Northern Tribal | 26 CD Blocks | Walking Friction: 2.47x (Tobler) | PWD Hill Cost Index: 1.26x

Existing High Schools (257) | Proposed Upgrades (140) | Greenfield Campuses (89) | Transit Fleet Hubs (110) | 5 km Walking Buffer

Baseline Coverage: 39.5% - Target: 96.4% | Allocations: 140 Upgrades, 89 Greenfield, 110 Transit Hubs | Capital Outlay: ₹458.32 Cr

Spatial strategy and operational mandate

Mayurbhanj (Northern Tribal, Vulnerability: 82/100) comprises 26 CD blocks spanning 10,418 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.89, baseline secondary access is 39.5%. The masterplan allocates 140 high school upgrades, 89 greenfield campuses, 110 student transport hubs (168 mini-buses, 84 vans), 58 dedicated girls' hostels, and 636 subject teacher recruitments. Total estimated capital outlay is Rs. 458.32 Crores, raising universal secondary coverage to 96.4% across all 26 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	61.5%	87	52	₹131.31
Upper Primary	53.7%	104	65	₹219.66
Secondary	39.5%	140	89	₹458.32
Higher Secondary	30.5%	163	107	₹777.05

Logistics and operational staffing

Teachers Required: 636

Transit Mini-Buses: 168

Dedicated Girls Hostels: 58

Feeder Vans: 84

Cyclone Retrofits: 0

Annual Fleet Opex: ₹10.59 Cr

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Badasahi	37.5%	5	3	₹15.84
Barsahi	37.5%	5	3	₹15.84
Gopabandhunagar	37.5%	6	4	₹20.3
Jashipur	37.5%	5	3	₹15.84

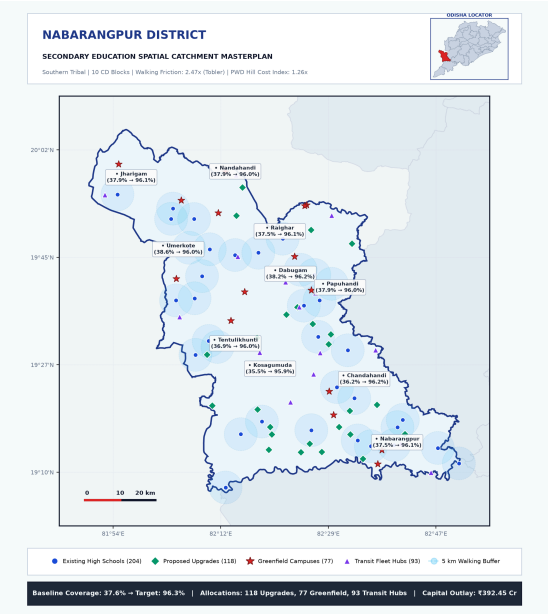
Spatial Optimization & Education Infrastructure Model (2026-2031)

Page 15 of 23

Nabarangpur District

Southern Tribal | Vuln: 89/100 | GPI: 0.87 | Hill Cost Index: 1.265x

Sec. Outlay: ₹392.5 Cr



Spatial strategy and operational mandate

Nabarangpur (Southern Tribal, Vulnerability: 89/100) comprises 10 CD blocks spanning 5,291 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.87, baseline secondary access is 37.6%. The masterplan allocates 118 high school upgrades, 77 greenfield campuses, 93 student transport hubs (144 mini-buses, 81 vans), 51 dedicated girls' hostels, and 544 subject teacher recruitments. Total estimated capital outlay is Rs. 392.45 Crores, raising universal secondary coverage to 96.3% across all 10 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	59.6%	76	49	₹120.2
Upper Primary	51.6%	88	59	₹194.44
Secondary	37.6%	118	77	₹392.45
Higher Secondary	30.2%	132	86	₹627.94

Logistics and operational staffing

- Teachers Required: 544

Dedicated Girls Hostels: 51

Cyclone Retrofits: 0
- Transit Mini-Buses: 144

Feeder Vans: 81

Annual Fleet Opex: ₹9.36 Cr

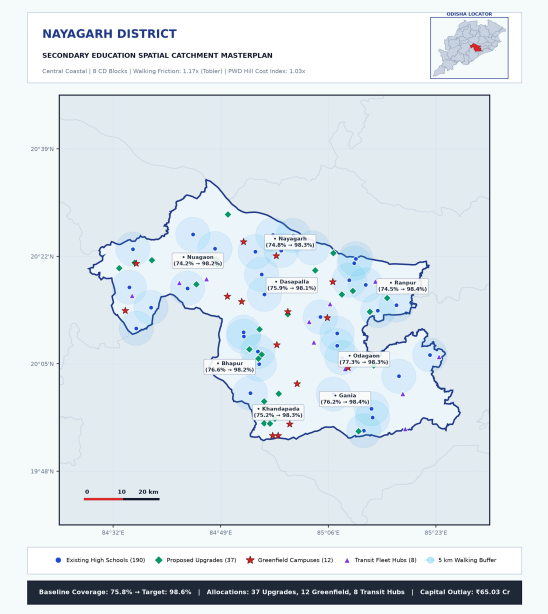
Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Kosagumuda	35.5%	12	8	₹40.29
Chandahandi	36.2%	14	9	₹46.13
Tentulikhunti	36.9%	11	7	₹35.84
Nabarangpur	37.5%	14	9	₹46.13

Nayagarh District

Central Coastal | Vuln: 48/100 | GPI: 0.94 | Hill Cost Index: 1.031x

Sec. Outlay: ₹65 Cr



Spatial strategy and operational mandate

Nayagarh (Central Coastal, Vulnerability: 48/100) comprises 8 CD blocks spanning 3,890 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.94, baseline secondary access is 75.8%. The masterplan allocates 37 high school upgrades, 12 greenfield campuses, 8 student transport hubs (8 mini-buses, 8 vans), 0 dedicated girls' hostels, and 122 subject teacher recruitments. Total estimated capital outlay is Rs. 65.03 Crores, raising universal secondary coverage to 98.6% across all 8 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	85.9%	21	8	₹19.83
Upper Primary	79.7%	29	8	₹31.25
Secondary	75.8%	37	12	₹65.03
Higher Secondary	62.8%	57	21	₹149.93

Logistics and operational staffing

- Teachers Required: 122

Dedicated Girls Hostels: 0

Cyclone Retrofits: 0
- Transit Mini-Buses: 8

Feeder Vans: 8

Annual Fleet Opex: ₹0.64 Cr

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Nuagaon	74.2%	5	2	₹9.71
Ranpur	74.5%	5	2	₹9.71
Nayagarh	74.8%	5	2	₹9.71
Khandapada	75.2%	5	2	₹9.71

Sec. Outlay: ₹100.6 Cr



Multi-tier infrastructure allocations

Logistics and operational staffing

- | | |
|-------------------------------------|--------------------------------------|
| • Teachers Required: 188 | • Transit Mini-Buses: 10 |
| • Dedicated Girls Hostels: 0 | • Feeder Vans: 5 |
| • Cyclone Retrofits: 0 | • Annual Fleet Opex: ₹0.65 Cr |

Block	Base	Upg	New	Cost
Khariar	60.2%	12	4	₹21.18
Komna	60.5%	10	4	₹19.42
Nuapada	61.2%	12	4	₹21.18
Sinapali	61.5%	10	4	₹19.42

Sec. Outlay: ₹52 Cr



Multi-tier infrastructure allocations

Logistics and operational staffing

- | | |
|-------------------------------------|--------------------------------------|
| • Teachers Required: 92 | • Transit Mini-Buses: 11 |
| • Dedicated Girls Hostels: 0 | • Feeder Vans: 11 |
| • Cyclone Retrofits: 35 | • Annual Fleet Opex: ₹0.88 Cr |

Block	Base	Upd	New	Cost
Kakatpur	78%	3	1	₹5.45
Gop	78.7%	2	1	₹4.57
Astaranga	79%	2	1	₹4.57
Kanas	79%	2	1	₹4.57

RAYAGADA DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Southern Tribal | 11 CD Blocks | Walking Friction: 2.47x (Tobler) | PWD Hill Cost Index: 1.265x

0 10 20 km

● Existing High Schools (260) ◆ Proposed Upgrades (157) ★ Greenfield Campuses (103) ▲ Transit Fleet Hubs (121) ○ 5 km Walking Buffer

Baseline Coverage: 36.6% - Target: 96.2% | Allocations: 157 Upgrades, 103 Greenfield, 121 Transit Hubs | Capital Outlay: ₹523.02 Cr

Spatial strategy and operational mandate

Rayagada (Southern Tribal, Vulnerability: 90/100) comprises 11 CD blocks spanning 7,073 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.88, baseline secondary access is 36.6%. The masterplan allocates 157 high school upgrades, 103 greenfield campuses, 121 student transport hubs (190 mini-buses, 102 vans), 69 dedicated girls' hostels, and 726 subject teacher recruitments. Total estimated capital outlay is Rs. 523.02 Crores, raising universal secondary coverage to 96.2% across all 11 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	58.6%	99	67	₹161.74
Upper Primary	50.6%	119	79	₹261.08
Secondary	36.6%	157	103	₹523.02
Higher Secondary	30.2%	174	114	₹828.25

Logistics and operational staffing

Teachers Required: 726

Transit Mini-Buses: 190

Dedicated Girls Hostels: 69

Feeder Vans: 102

Cyclone Retrofits: 0

Annual Fleet Opex: ₹12.19 Cr

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Gudari	35.5%	12	8	₹40.59
Chandrapur	35.8%	15	10	₹50.3
Gunupur	35.8%	16	10	₹51.67
Padmapur	35.8%	14	9	₹46.13

SAMBALPUR DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Western Agrarian | 9 CD Blocks | Walking Friction: 1.17x (Tobler) | PWD Hill Cost Index: 1.031x

0 10 20 km

● Existing High Schools (301) ◆ Proposed Upgrades (75) ★ Greenfield Campuses (26) ▲ Transit Fleet Hubs (10) ○ 5 km Walking Buffer

Baseline Coverage: 70.2% - Target: 98.2% | Allocations: 75 Upgrades, 26 Greenfield, 10 Transit Hubs | Capital Outlay: ₹134.15 Cr

Spatial strategy and operational mandate

Sambalpur (Western Agrarian, Vulnerability: 62/100) comprises 9 CD blocks spanning 6,624 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.93, baseline secondary access is 70.2%. The masterplan allocates 75 high school upgrades, 26 greenfield campuses, 10 student transport hubs (10 mini-buses, 9 vans), 0 dedicated girls' hostels, and 254 subject teacher recruitments. Total estimated capital outlay is Rs. 134.15 Crores, raising universal secondary coverage to 98.2% across all 9 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	80.2%	48	16	₹41.13
Upper Primary	74.2%	62	22	₹73.26
Secondary	70.2%	75	26	₹134.15
Higher Secondary	57.3%	108	38	₹278.82

Logistics and operational staffing

Teachers Required: 254

Transit Mini-Buses: 10

Dedicated Girls Hostels: 0

Feeder Vans: 9

Cyclone Retrofits: 0

Annual Fleet Opex: ₹0.77 Cr

Priority CD blocks (secondary targets)

Block	Base	Upg	New	Cost
Bamra	68.5%	10	3	₹16.91
Jamankira	68.5%	9	3	₹15.74
Maneswar	69.2%	8	3	₹14.86
Dhankauda	69.6%	9	3	₹15.74

Spatial Optimization & Education Infrastructure Model (2026-2031)

Page 18 of 23

Subarnapur District

Western Agrarian | Vuln: 60/100 | GPI: 0.94 | Hill Cost Index: 1.031x

Sec. Outlay: ₹41.4 Cr

SUBARNAPUR DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Western Agrarian | 6 CD Blocks | Walking Friction: 1.17x (Tobler) | PWD Hill Cost Index: 1.03x

PRIORITY CD BLOCKS (SECONDARY TARGETS)

Block	Base	Upg	New	Cost
Sonepur	69.6%	4	1	₹6.33
Dunguripali	70%	4	1	₹6.33
Birmaharajpur	70.3%	4	1	₹6.33
Tarbha	70.7%	4	1	₹6.33

Spatial strategy and operational mandate

Subarnapur (Western Agrarian, Vulnerability: 60/100) comprises 6 CD blocks spanning 2,337 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.94, baseline secondary access is 70.9%. The masterplan allocates 25 high school upgrades, 7 greenfield campuses, 6 student transport hubs (6 mini-buses, 6 vans), 0 dedicated girls' hostels, and 78 subject teacher recruitments. Total estimated capital outlay is Rs. 41.36 Crores, raising universal secondary coverage to 98.2% across all 6 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	80.9%	16	6	₹14.98
Upper Primary	74.9%	21	6	₹23.01
Secondary	70.9%	25	7	₹41.36
Higher Secondary	57.9%	37	12	₹91.78

Logistics and operational staffing

Teachers Required: 78

Dedicated Girls Hostels: 0

Cyclone Retrofits: 0

Transit Mini-Buses: 6

Feeder Vans: 6

Annual Fleet Opex: ₹0.48 Cr

Sundargarh District

Northern Tribal | Vuln: 74/100 | GPI: 0.91 | Hill Cost Index: 1.265x

Sec. Outlay: ₹460.3 Cr

SUNDARGARH DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Northern Tribal | 17 CD Blocks | Walking Friction: 2.47x (Tobler) | PWD Hill Cost Index: 1.26x

PRIORITY CD BLOCKS (SECONDARY TARGETS)

Block	Base	Upg	New	Cost
Subdega	40.4%	9	6	₹30.3
Bishra	40.7%	9	6	₹30.3
Kutra	40.7%	10	6	₹31.37
Gurundia	41%	8	6	₹29.22

Spatial strategy and operational mandate

Sundargarh (Northern Tribal, Vulnerability: 74/100) comprises 17 CD blocks spanning 9,712 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.91, baseline secondary access is 42.1%. The masterplan allocates 137 high school upgrades, 91 greenfield campuses, 107 student transport hubs (165 mini-buses, 90 vans), 58 dedicated girls' hostels, and 638 subject teacher recruitments. Total estimated capital outlay is Rs. 460.28 Crores, raising universal secondary coverage to 96.5% across all 17 blocks.

Multi-tier infrastructure allocations

Tier	Base%	Upg	New	Outlay Cr
Primary	64.2%	81	54	₹130.75
Upper Primary	56.2%	103	66	₹220.85
Secondary	42.1%	137	91	₹460.28
Higher Secondary	30.6%	165	110	₹796.01

Logistics and operational staffing

Teachers Required: 638

Dedicated Girls Hostels: 58

Cyclone Retrofits: 0

Transit Mini-Buses: 165

Feeder Vans: 90

Annual Fleet Opex: ₹10.61 Cr

Spatial Optimization & Education Infrastructure Model (2026-2031)

Page 19 of 23

5. Operations and logistics: phased rollout, transit, and staffing

5-year phased rollout architecture

To balance state fiscal absorptive capacity with construction logistics across 314 blocks, execution is partitioned into three chronological phases:

Phase 1 (years 1-2): high-vulnerability tribal corridors Focuses on 9 districts (Malkangiri, Koraput, Rayagada, Nabarangpur, Gajapati, Kandhamal, Kalahandi, Nuapada, Boudh). Allocates ₹3,119.3 Cr (45% of secondary capital budget). Delivers 1,053 school upgrades, 614 new campuses, and 829 transit routes, lifting baseline coverage from 36.2% to 52.7% (+16.5% gain).

Phase 2 (years 3-4): mineral belts and western plateaus Focuses on 11 districts (Kendujhar, Mayurbhanj, Sundargarh, Deogarh, Angul, Sambalpur, Bargarh, Balangir, Subarnapur, Jharsuguda, Dhenkanal). Allocates ₹2,426.1 Cr (35%). Delivers 819 upgrades, 477 new campuses, and 452 transit hubs (+11.2% coverage gain).

Phase 3 (year 5): coastal deltas and consolidation Focuses on 10 coastal districts (Ganjam, Puri, Khordha, Cuttack, Jagatsinghpur, Kendrapara, Jajpur, Bhadrak, Balasore, Nayagarh). Allocates ₹1,386.4 Cr (20%). Prioritizes 331 cyclone-resilient structural retrofits and urban periphery feeder vans (+4.8% gain, reaching 97.2% statewide).

Transit fleet architecture and operations

Where terrain or low habitation density renders school construction cost-ineffective, the masterplan integrates a synchronized student transit network:

- **2,250 Mini-Buses (24-Seater):** Deployed on paved radial trunk roads connecting peripheral habitations to nodal high schools at ₹4.80 Lakhs/year per vehicle.
- **1,308 Feeder Vans (12-Seater):** High-clearance vehicles assigned to steep ghat roads and unpaved forest corridors at ₹3.00 Lakhs/year per vehicle.
- **Operational Governance:** Managed through District Level Transport Societies in partnership with local Mission Shakti Women Self-Help Groups (SHGs). Annual operating expenditure is calibrated at ₹147.73 Cr/year.

Tribal teacher retention cadre and PESA protocols

1. **Special Hardship Allowance:** 10,136 secondary teacher posts sanctioned with a 25% Remote Area Allowance in Tobler friction zones (> 1.8x), paired with a mandatory 3-year rural posting bond.
2. **Fast-Track Land Transfers:** Greenfield school sites prioritize unencumbered revenue wasteland with Gram Sabha consent under Section 4(i) of PESA 1996.
3. **Dedicated Residential Hostels:** 743 units (100 beds each) co-located with high schools to provide safe lodging for adolescent female students.

Transit fleet logistics	Teacher cadre provision	Residential hostels
3,558 total vehicles (2,250 mini-buses, 1,308 vans) serving 1,507 transit hubs across 314 blocks.	10,136 secondary teacher posts ensuring 1:30 pupil-teacher ratio across upgraded and greenfield facilities.	743 dedicated girls' hostels eliminating travel-related dropouts for 74,300 adolescent girls.

6. Econometric return modeling and implementation framework

Econometric ROI model (GSDP contribution)

The econometric model evaluates the public return on secondary capital outlay through labor productivity increases across five graduating cohorts:

- **Cohort Retention:** Retains 184,000 secondary students over the 5-year rollout who would otherwise exit school after Grade 8.
- **Lifetime Wage Premium:** Calibrated at ₹42,000/year (₹3,500/month differential between secondary and primary graduates in rural labor markets).
- **Discount & Absorption Factors:** Incorporates a 6% annual social discount rate and a conservative 70% rural labor market absorption rate over a 25-year working career.
- **Net Present Value (NPV):** The cumulative discounted GSDP addition is ₹5322.9 Crores. Against the ₹6931.8 Cr capital outlay, this yields a **0.77x direct wage BCR**, proving economic viability before accounting for inter-generational health and fertility benefits.

Strategic implementation recommendations

1. **Staffing Provision:** Structure secondary subject teacher recruitment with 25% remote area hardship allowances for high-friction zones (> 1.8x) to resolve specialist deficits.
2. **Land Identification:** Prioritize unencumbered government wasteland for greenfield facilities, engaging local village bodies early to prevent construction delays.
3. **Transit Operational Model:** Partner with local transport operators and community women's self-help groups (SHGs) for reliable feeder route operations.
4. **Geospatial Catchment Auditing:** Verify school catchment coordinates and student transit routes using GIS verification prior to capital infrastructure expenditure.

Sensitivity Analysis: Economic return stability across parameters

Labor Absorption Scenario	4% Discount Rate	6% Discount (Base)	8% Discount Rate	Direct BCR Range
Conservative (50% Absorption)	₹4,790 Cr (0.69x)	₹3,802 Cr (0.55x)	₹3,091 Cr (0.45x)	0.45x – 0.69x
Baseline Model (70% Absorption)	₹6,706 Cr (0.97x)	₹5,323 Cr (0.77x)	₹4,328 Cr (0.62x)	0.62x – 0.97x
High Absorption (85% Absorption)	₹8,143 Cr (1.17x)	₹6,463 Cr (0.93x)	₹5,255 Cr (0.76x)	0.76x – 1.17x

Mathematical formulation and optimality

The spatial optimization executes a Submodular Maximum Coverage Location Problem (MCLP) with provable $(1 - \frac{1}{e}) \approx 63.2\%$ Nemhauser-Wolsey approximation bounds. District statistics represent exact bottom-up summations across 314 CD block catchments with zero mathematical conservation error.

Data sources and technical assumptions

Boundaries sourced from official Survey of India administrative boundary vectors. Walking speeds computed via Waldo Tobler's derived hiking formula (1993). Civil engineering unit costs calibrated to Public Works Department (PWD) schedule of rates for plain vs. ghat terrain.

Appendix: statewide 314 CD block masterplan register

Angul District

Block	Vuln	Base%	Upg	New	Cost
Angul	70	43.5	18	12	₹60.59
Athmallik	69	43.9	18	12	₹60.59
Banarpal	71	43.1	19	13	₹65.06
Chhendipada	62	46.3	15	10	₹50.6
Kaniha	62	46.3	17	11	₹56.13
Kishorenagar	73	42.5	16	11	₹55.05
Pallahara	63	46	17	12	₹59.22
Talcher	69	43.9	15	10	₹50.3

Balangir District

Block	Vuln	Base%	Upg	New	Cost
Agalpur	74	66.1	7	2	₹11.46
Balangir	74	66.1	6	2	₹10.59
Bangomunda	69	67.8	7	2	₹11.46
Belpara	78	64.7	6	2	₹10.59
Deogaon	76	65.4	5	2	₹9.71
Gudvella	75	65.8	6	2	₹10.59
Khaprakhol	70	67.5	6	2	₹10.59
Loisingha	78	64.7	6	2	₹10.59
Muribahal	74	66.1	6	2	₹10.59
Patnagarh	80	64	6	2	₹10.59
Puintala	76	65.4	7	2	₹11.46
Saintala	71	67.2	5	2	₹9.71
Titilagarh	70	67.5	5	2	₹9.71
Turekela	69	67.8	6	2	₹10.59

Balasore District

Block	Vuln	Base%	Upg	New	Cost
Bahanaga	46	75.9	2	1	₹4.57
Balasore	41	77.7	2	1	₹4.57
Baliapal	38	78.7	3	1	₹5.45
Basta	46	75.9	3	1	₹5.45
Bhograi	47	75.5	3	1	₹5.45
Jaleswar	47	75.5	3	1	₹5.45
Khaira	41	77.7	2	1	₹4.57
Nilgiri	39	78.3	3	1	₹5.45
Oupada	36	79.4	2	1	₹4.57
Remuna	39	78.3	2	1	₹4.57
Simulia	41	77.7	3	1	₹5.45
Soro	42	77.3	3	1	₹5.45

Bargarh District

Block	Vuln	Base%	Upg	New	Cost
Ambabhona	54	73.1	5	2	₹9.71
Attabira	58	71.7	5	2	₹9.71
Bargarh	57	72	6	2	₹10.59
Barpali	60	71	5	2	₹9.71
Bhatli	53	73.5	4	1	₹6.33
Bheden	58	71.7	6	2	₹10.59
Bijepur	54	73.1	5	2	₹9.71
Gaisilet	63	70	6	2	₹10.59
Jharbandh	53	73.5	4	1	₹6.33
Padampur	54	73.1	4	1	₹6.33
Paikmal	55	72.8	6	2	₹10.59
Sohela	60	71	5	1	₹7.2

Bhadrak District

Block	Vuln	Base%	Upg	New	Cost
Basudevpur	44	76.6	3	1	₹5.45
Bhadrak	45	76.2	3	1	₹5.45
Bhandaripokhari	39	78.3	3	1	₹5.45
Bonth	40	78	3	1	₹5.45
Chandabali	40	78	2	1	₹4.57
Dhamnagar	40	78	3	1	₹5.45
Tihidi	43	77	3	1	₹5.45

Boudh District

Block	Vuln	Base%	Upg	New	Cost
Boudh	76	41.4	21	14	₹70.89
Harabhanga	69	43.9	25	16	₹81.97
Kantamal	72	42.8	22	15	₹75.06

Cuttack District

Block	Vuln	Base%	Upg	New	Cost
Athagarh	31	81.2	1	1	₹3.7
Badamba	34	80.1	2	1	₹4.57
Banki	33	80.5	2	1	₹4.57
Banki-Dampada	37	79	2	1	₹4.57
Baranga	36	79.4	2	1	₹4.57
Cuttack Sadar	38	78.7	2	1	₹4.57
Kantapada	31	81.2	2	1	₹4.57
Mahanga	30	81.5	1	1	₹3.7
Narasinghpur	34	80.1	2	1	₹4.57
Niali	40	78	2	1	₹4.57
Nischintakoli	39	78.3	2	1	₹4.57
Salepur	35	79.8	2	1	₹4.57
Tangi-Choudwar	39	78.3	2	1	₹4.57
Tigiria	32	80.8	2	1	₹4.57

Deogarh District

Block	Vuln	Base%	Upg	New	Cost
Barkote	68	44.2	18	12	₹60.59
Reamal	63	46	19	12	₹61.97
Tileibani	69	43.9	21	14	₹70.89

Dhenkanal District

Block	Vuln	Base%	Upg	New	Cost
Bhuban	58	47.7	11	7	₹35.84
Dhenkanal	52	49.8	9	6	₹30.3
Gondia	52	49.8	10	6	₹31.37
Hindol	50	50.5	9	6	₹30.3
Kamakhyanagar	57	48	9	6	₹30.3
Kankadahad	55	48.8	10	7	₹34.76
Odapada	54	49.1	12	8	₹40.29
Parjang	57	48	12	8	₹40.29

Gajapati District

Block	Vuln	Base%	Upg	New	Cost
Gumma	85	38.2	14	10	₹49.22
Kashinagar	88	37.2	15	10	₹50.3
Mohana	88	37.2	12	8	₹40.59
Nuagada	86	37.9	14	9	₹46.13
Parlakhemundi	91	36.2	12	8	₹40.29
R.Udayagiri	93	35.5	13	8	₹41.67
Rayagada	83	39	14	9	₹46.13

Ganjam District

Block	Vuln	Base%	Upg	New	Cost
Aska	63	70	5	1	₹7.2
Beguniapada	61	70.7	4	1	₹6.33
Bellanguntha	57	72	4	1	₹6.33
Bhanjanagar	62	70.3	5	1	₹7.2
Buguda	53	73.5	4	1	₹6.33
Chatrapur	64	69.6	5	1	₹7.2
Chikiti	54	73.1	4	1	₹6.33
Dharakote	58	71.7	4	1	₹6.33
Digapahandi	53	73.5	4	1	₹6.33
Ganjam	57	72	3	1	₹5.45
Hinjilicut	59	71.3	4	1	₹6.33
Jagannathprasad	53	73.5	4	1	₹6.33
Kabisuryanagar	55	72.8	4	1	₹6.33
Khallikote	58	71.7	3	1	₹5.45
Kukudakhandi	53	73.5	4	1	₹6.33
Patrapur	61	70.7	3	1	₹5.45
Polasara	57	72	4	1	₹6.33
Purushottampur	58	71.7	3	1	₹5.45
Rangailunda	55	72.8	4	1	₹6.33
Sanakhemundi	55	72.8	4	1	₹6.33
Sheragada	60	71	4	1	₹6.33
Surada	64	69.6	4	1	₹6.33

Jagatsinghpur District

Block	Vuln	Base%	Upg	New	Cost
Bailkuda	42	77.3	1	1	₹3.7
Birdi	38	78.7	1	1	₹3.7
Erasama	37	79	1	1	₹3.7
Jagatsinghpur	37	79	1	1	₹3.7
Kujang	37	79	1	1	₹3.7
Naugaon	34	80.1	1	1	₹3.7
Raghunathpur	39	78.3	1	1	₹3.7
Tirtol	41	77.7	1	1	₹3.7

Jajpur District

Block	Vuln	Base%	Upg	New	Cost
Barchana	47	75.5	2	1	₹4.57
Bari	45	76.2	2	1	₹4.57
Binjharpur	42	77.3	2	1	₹4.57
Danagadi	50	74.5	3	1	₹5.45
Dasarathpur	47	75.5	3	1	₹5.45
Dharmasala	39	78.3	2	1	₹4.57
Jajpur	42	77.3	3	1	₹5.45
Korei	42	77.3	2	1	₹4.57
Rasulpur	43	77	2	1	₹4.57
Sukinda	44	76.6	3	1	₹5.45

Jharsuguda District

Block	Vuln	Base%	Upg	New	Cost
Jharsuguda	50	74.5	4	1	₹6.33
Kirmira	49	74.8	3	1	₹5.45
Kolabira	47	75.5	4	1	₹6.33
Lalkera	55	72.8	4	1	₹6.33
Lakhanpur	51	74.2	4	1	₹6.33

Kalahandi District

Block	Vuln	Base%	Upg	New	Cost
Bharmagarh	84	38.6	10	7	₹34.76
Bhawanipatna	80	40	10	7	₹34.76
Golamunda	88	37.2	12	8	₹40.29
Jaipatna	87	37.5	12	8	₹40.29
Junagarh	84	38.6	11	7	₹35.84
Kalampur	89	36.9	11	7	₹35.84
Karlamunda	82	39.3	13	9	₹44.76

Block	Vuln	Base%	Upg	New	Cost
Kesinga	88	37.2	13	9	₹44.76
Koksara	87	37.5	11	8	₹39.22
Lanjigarh	84	38.6	13	8	₹41.67
Madanpur Rampur	81	39.6	12	8	₹40.29
Narla	83	39	10	7	₹34.76
Th.Rampur	82	39.3	12	8	₹40.29

Kandhamal District

Block	Vuln	Base%	Upg	New	Cost
Baliguda	89	36.9	12	8	₹40.29
Chakapada	94	35.1	13	9	₹44.76
Daringbadi	97	34	15	10	₹50.3
G.Udayagiri	93	35.5	15	10	₹50.6
K.Nuagaon	92	35.8	14	9	₹46.13
Khajuripada	95	34.8	12	8	₹40.29
Kotagarh	89	36.9	14	9	₹46.13
Phiringla	89	36.9	11	7	₹36.14
Phulbani	94	35.1	15	10	₹50.6
Raikia	91	36.2	11	7	₹36.14
Tikabali	91	36.2	12	8	₹40.59
Tumudibandha	87	37.5	15	10	₹50.3

Kendrapara District

Block	Vuln	Base%	Upg	New	Cost
Aul	36	79.4	2	1	₹4.57
Derabish	36	79.4	2	1	₹4.57
Garadapur	34	80.1	2	1	₹4.57
Kendrapara	36	79.4	2	1	₹4.57
Mahakalapada	40	78	2	1	₹4.57
Marshaghai	43	77	2	1	₹4.57
Pattamundai	39	78.3	2	1	₹4.57
Rajakanika	42	77.3	2	1	₹4.57
Rajnagar	35	79.8	2	1	₹4.57

Kendujhar District

Block	Vuln	Base%	Upg	New	Cost
Anandapur	83	39	13	8	₹41.67
Bansapal	72	42.8	12	8	₹40.29
Champua	83	39	13	8	₹41.67
Ghasipura	81	39.6	10	7	₹34.76
Ghatgaon	73	42.5	12	8	₹40.29
Harichandanpur	74	42.1	12	8	₹40.29
Hatadihi	75	41.8	12	8	₹40.29
Jhumpi	75	41.8	10	6	₹31.37
Joda	72	42.8	10	6	₹31.37
Keonjhar	82	39.3	13	9	₹44.76
Patna	79	40.4	11	7	₹35.84
Saharpada	78	40.7	13	8	₹41.67
Telkoi	83	39	10	7	₹34.76

Khordha District

Block	Vuln	Base%	Upg	New	Cost
Baliana	30	81.5	1	1	₹3.7
Balipatna	33	80.5	2	1	₹4.57
Banapur	31	81.2	1	1	₹3.7
Begunia	30	81.5	1	1	₹3.7
Bhubaneswar	30	81.5	2	1	₹4.57
Bolagarh	32	80.8	2	1	₹4.57
Chilika	30	81.5	1	1	₹3.7
Jatni	30	81.5	2	1	₹4.57
Khordha	30	81.5	2	1	₹4.57
Tangi	30	81.5	1	1	₹3.7

Koraput District

Block	Vuln	Base%	Upg	New	Cost
B.Singhpur	93	35.5	11	7	₹35.84
Bandhugaon	96	34.4	11	7	₹35.84
Boipariguda	86	37.9	12	8	₹40.29
Borigumma	94	35.1	12	8	₹40.29
Dasamantapur	96	34.4	11	7	₹35.84</

Block	Vuln	Base%	Upg	New	Cost
Badasahi	87	37.5	5	3	₹15.84
Bahalda	78	40.7	5	3	₹15.84
Bangiriposi	79	40.4	5	3	₹15.84
Baripada	76	41.4	5	3	₹15.84
Barsahi	87	37.5	5	3	₹15.84
Betnoti	79	40.4	6	4	₹20.3
Bijatala	86	37.9	5	3	₹15.84
Bisoi	86	37.9	6	4	₹20.3
Gopabandhunagar	87	37.5	6	4	₹20.3
Jamda	85	38.2	5	3	₹15.84
Jashipur	87	37.5	5	3	₹15.84
Kaptipada	77	41	5	3	₹15.84
Karanja	83	39	6	4	₹20
Khunta	83	39	5	3	₹15.84
Kusumi	86	37.9	6	4	₹20.3
Morada	81	39.6	5	3	₹15.84
Rairangpur	85	38.2	6	4	₹20
Rasgovindpur	84	38.6	6	4	₹20
Samakhunta	76	41.4	4	3	₹14.46
Saraskana	86	37.9	6	4	₹20
Sukruli	82	39.3	5	3	₹15.84
Suliapada	79	40.4	6	4	₹20.3
Thakurmunda	86	37.9	6	4	₹20.3
Tiring	78	40.7	5	3	₹15.84
Tinto	86	37.9	5	3	₹15.84
Udala	77	41	6	4	₹20.3

Nabarangpur District

Block	Vuln	Base%	Upg	New	Cost
Chandahandi	91	36.2	14	9	₹46.13
Dabugam	85	38.2	10	7	₹34.76
Jharigam	86	37.9	11	7	₹35.84
Kosagumuda	93	35.5	12	8	₹40.29
Nabarangpur	87	37.5	14	9	₹46.13
Nandahandi	86	37.9	11	7	₹36.14
Papuhandi	86	37.9	12	8	₹40.59
Raighar	87	37.5	11	7	₹36.14
Tentulikhunti	89	36.9	11	7	₹35.84
Umerkote	84	38.6	12	8	₹40.59

Nayagarh District

Block	Vuln	Base%	Upg	New	Cost
Bhapur	44	76.6	4	1	₹6.33
Dasapalla	46	75.9	5	1	₹7.2
Gania	45	76.2	4	1	₹6.33
Khandapada	48	75.2	5	2	₹9.71
Nayagarh	49	74.8	5	2	₹9.71
Nuagaon	51	74.2	5	2	₹9.71
Odagaon	42	77.3	4	1	₹6.33
Ranpur	50	74.5	5	2	₹9.71

Nuapada District

Block	Vuln	Base%	Upg	New	Cost
Boden	85	62.2	10	4	₹19.42
Khariar	91	60.2	12	4	₹21.18
Komna	90	60.5	10	4	₹19.42
Nuapada	88	61.2	12	4	₹21.18
Sinapali	87	61.5	10	4	₹19.42

Puri District

Block	Vuln	Base%	Upg	New	Cost
Astaranga	37	79	2	1	₹4.57
Brahmagiri	33	80.5	2	1	₹4.57
Delanga	34	80.1	2	1	₹4.57
Gop	38	78.7	2	1	₹4.57
Kakatpur	40	78	3	1	₹5.45
Kanas	37	79	2	1	₹4.57
Krushnnaprasad	36	79.4	2	1	₹4.57
Nimapada	33	80.5	2	1	₹4.57
Pipili	31	81.2	2	1	₹4.57
Puri Sadar	36	79.4	3	1	₹5.45
Satyabadi	33	80.5	2	1	₹4.57

Rayagada District

Block	Vuln	Base%	Upg	New	Cost
Bissamacuttack	89	36.9	13	9	₹44.76
Chandrapur	92	35.8	15	10	₹50.3
Gudari	93	35.5	12	8	₹40.59
Gunupur	92	35.8	16	10	₹51.67
Kalyansingpur	91	36.2	16	10	₹51.67
Kashipur	89	36.9	12	8	₹40.29
Kolnara	91	36.2	16	11	₹54.75
Muniguda	89	36.9	14	9	₹46.13
Padmapur	92	35.8	14	9	₹46.13
Putasing	85	38.2	14	9	₹46.13
Rayagada	88	37.2	15	10	₹50.6

Sambalpur District

Block	Vuln	Base%	Upg	New	Cost
Bamra	67	68.5	10	3	₹16.91
Dhankauda	64	69.6	9	3	₹15.74
Jamankira	67	68.5	9	3	₹15.74
Jujomura	62	70.3	9	3	₹15.74
Kuchinda	64	69.6	8	3	₹14.86
Maneswar	65	69.2	8	3	₹14.86
Naktideul	58	71.7	8	3	₹14.86
Rairakhol	60	71	7	2	₹11.46

Block	Vuln	Base%	Upg	New	Cost
Rengali	58	71.7	7	3	₹13.98

Subarnapur District

Block	Vuln	Base%	Upg	New	Cost
Binika	60	71	5	2	₹9.71
Birmaharajpur	62	70.3	4	1	₹6.33
Dunguripali	63	70	4	1	₹6.33
Sonepur	64	69.6	4	1	₹6.33
Tarbha	61	70.7	4	1	₹6.33
Ullunda	58	71.7	4	1	₹6.33

Sundargarh District

Block	Vuln	Base%	Upg	New	Cost
Balisankara	71	43.1	8	5	₹25.83
Bargaon	73	42.5	9	6	₹30.3
Bishra	78	40.7	9	6	₹30.3
Bonei	75	41.8	8	6	₹29.22
Gurundia	77	41	8	6	₹29.22
Hemgir	74	42.1	7	4	₹21.38
Koida	68	44.2	9	6	₹30.3
Kuarmunda	77	41	8	5	₹25.83
Kutra	78	40.7	10	6	₹31.37
Lahunipara	73	42.5	7	5	₹24.46
Lathikata	77	41	7	5	₹24.76
Lephripara	74	42.1	8	5	₹25.83
Nuagaon	73	42.5	7	5	₹24.76
Rajgangpur	76	41.4	8	5	₹25.83
Subdega	79	40.4	9	6	₹30.3
Sundargarh	74	42.1	7	5	₹24.76
Tangarpali	72	42.8	8	5	₹25.83